

Robotics Math - Assignment 02

> Suraj Dayma | U20230102

Task 1 - 4: Done is last HW

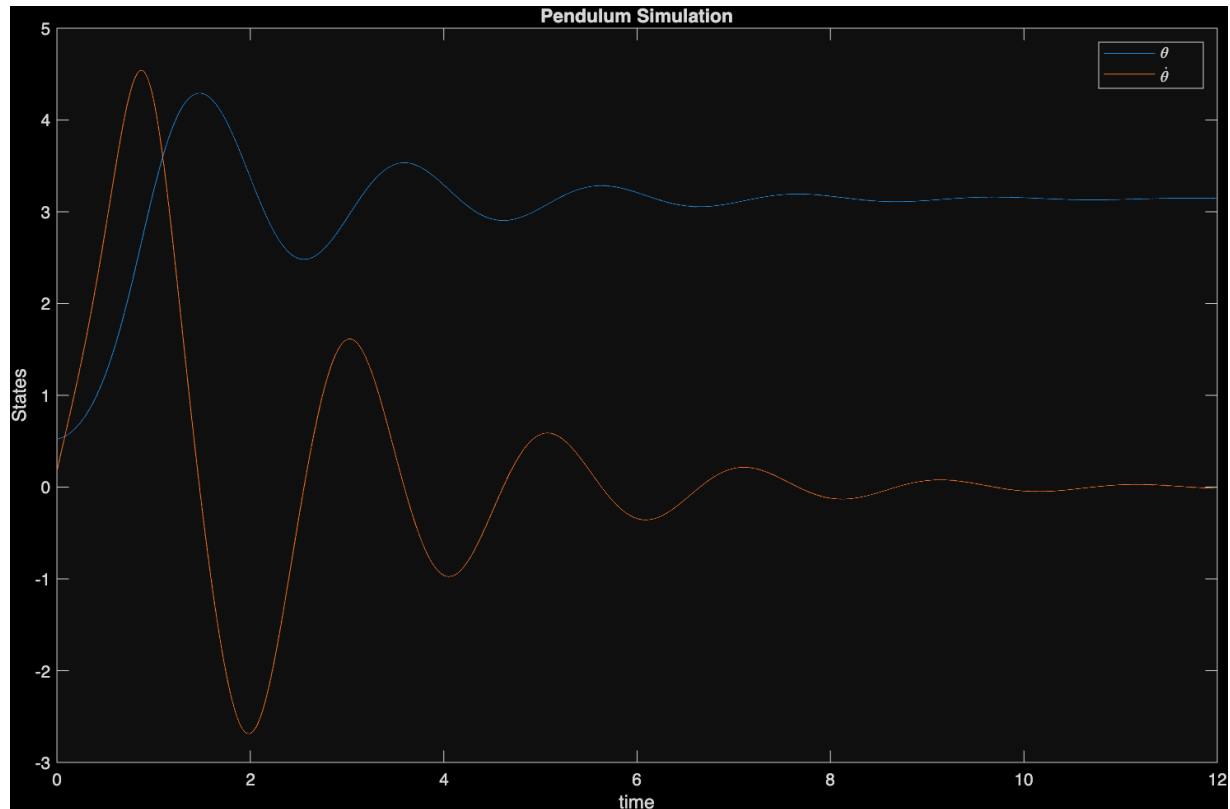
Task 5

5) Cook up, or find, some ODE, or set of first order ODEs, and solve them and explore something hopefully-interesting about the solution.

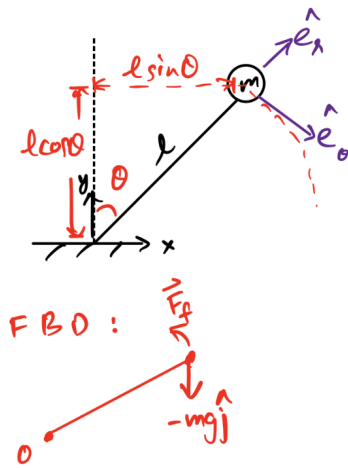
ODE 1 (Done last HW)

Done in file *interestingODE1.m*

The equations of motion of an inverted pendulum with air resistance were derived as shown below and the dynamics were simulated using Euler's method.



The dynamics of this is derived below.



$$\vec{r} = l \hat{e}_r, \quad \vec{v} = l\dot{\theta} \hat{e}_\theta$$

$$\vec{a} = l\ddot{\theta} \hat{e}_\theta - l\dot{\theta}^2 \hat{e}_r$$

$$\text{Air Friction } \vec{F}_f = -\gamma l \dot{\theta} \hat{v} = -\gamma l \dot{\theta} \hat{e}_\theta$$

Angular momentum balance,

$$\{ + \sum \vec{M}_O = \vec{r} \times m \vec{a} \} \cdot \hat{k}$$

$$\Rightarrow (\vec{r} \times \vec{\omega}) + \vec{r} \times (-\gamma l \dot{\theta}) \hat{e}_\theta = \vec{r} \times m(l\ddot{\theta} \hat{e}_\theta - l\dot{\theta}^2 \hat{e}_r)$$

$$\Rightarrow -\gamma l^2 \dot{\theta} + lmg \sin \theta = ml^2 \ddot{\theta}$$

states : $\theta, \dot{\theta}$

$$\dot{z}_1 = \dot{\theta} = z_2 = \dot{\theta}$$

$$\dot{z}_2 = \ddot{\theta} = \frac{1}{ml^2} \left(-\gamma l^2 z_2 + lmg \sin(z_1) \right)$$

ODE 2 - Airy Function

Done in file **interestingODE2.m**

I googled interesting ODEs and one of the results was the [Airy Function](#) which is $\frac{d^2 x}{dt^2} - tx = 0$ which is solved using the Euler method.

Airy Equation

$$\frac{d^2 x}{dt^2} - tx = 0$$

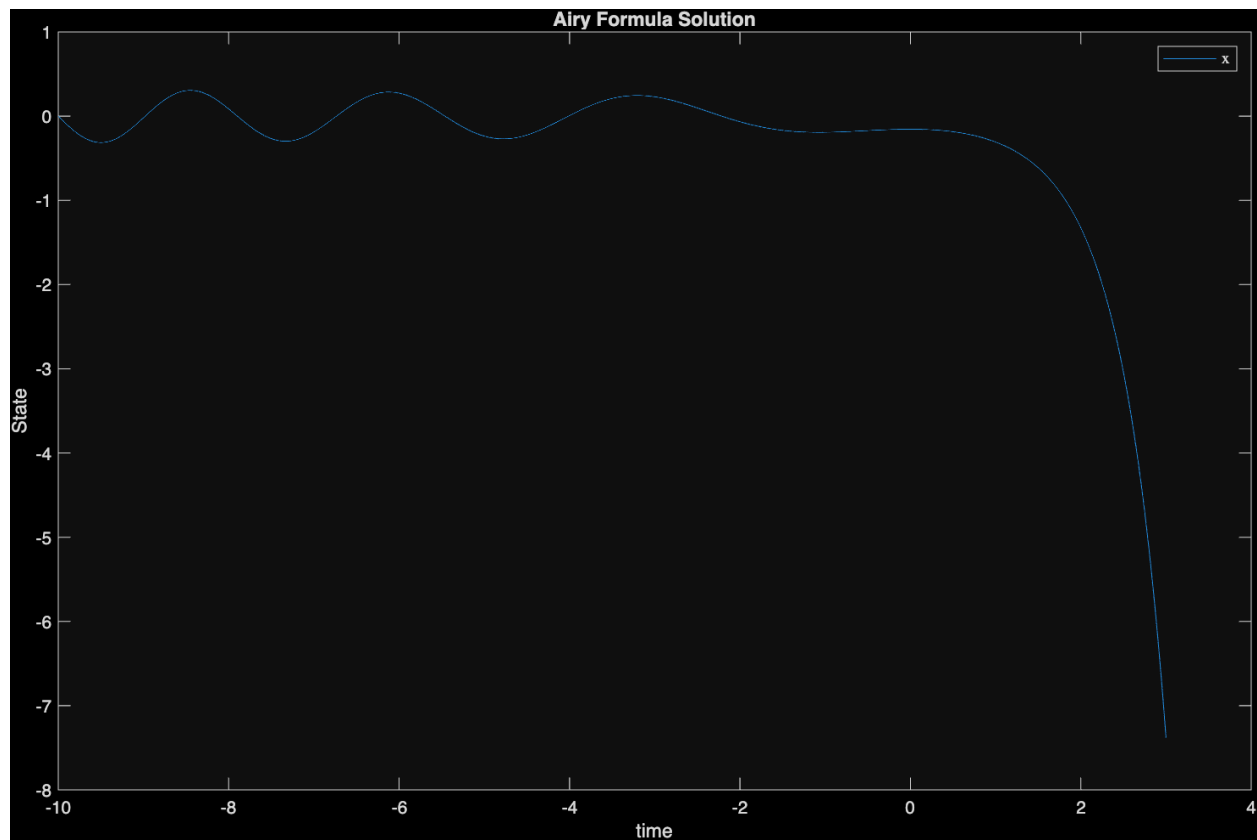
$$\frac{d^2 x}{dt^2} = tx$$

$$\frac{d}{dt} \frac{dx}{dt} = tx$$

let $\frac{d}{dt} x = x_2$

$$\begin{aligned} \dot{x}_2 &= tx \\ \dot{x}_1 &= x_2 \end{aligned}$$

The solution for $x_0 = 0.1$ is:



The interesting thing here is that the solution is oscillatory for $t < 0$ and exponential for $t > 0$.

ODE 3 - Van der Pol oscillator

Done in file [interestingODE3.m](#). Another result that came up was the [Van der Pol oscillator](#) with the ODE:

$$\frac{d^2x}{dt^2} - \mu(1 - x^2) \frac{dx}{dt} + x = 0$$

VDP Osc.

$$\frac{d^2x}{dt^2} - \mu(1 - x^2) \frac{dx}{dt} + x = 0$$

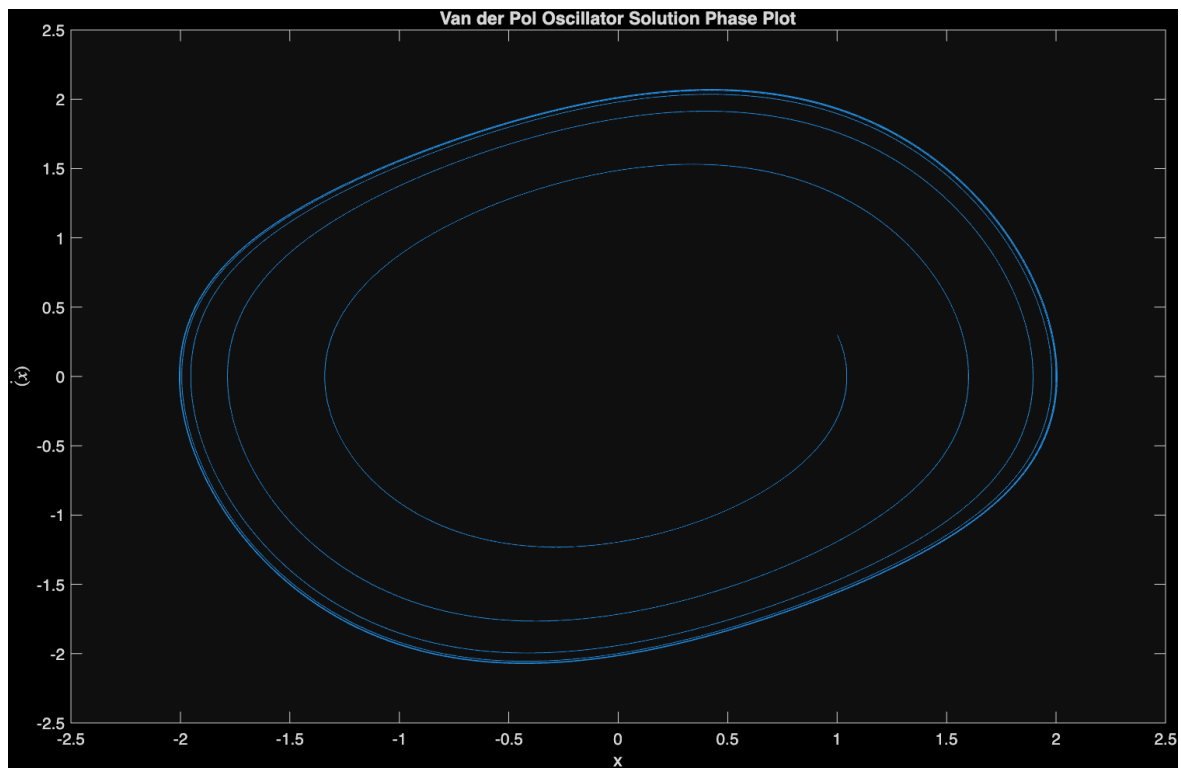
Let $x_1 = x$, $x_2 = \frac{dx}{dt} \Rightarrow \boxed{\dot{x}_1 = x_2}$

$\Rightarrow \dot{x}_2 - \mu(1 - x_1^2)x_2 + x_1 = 0$

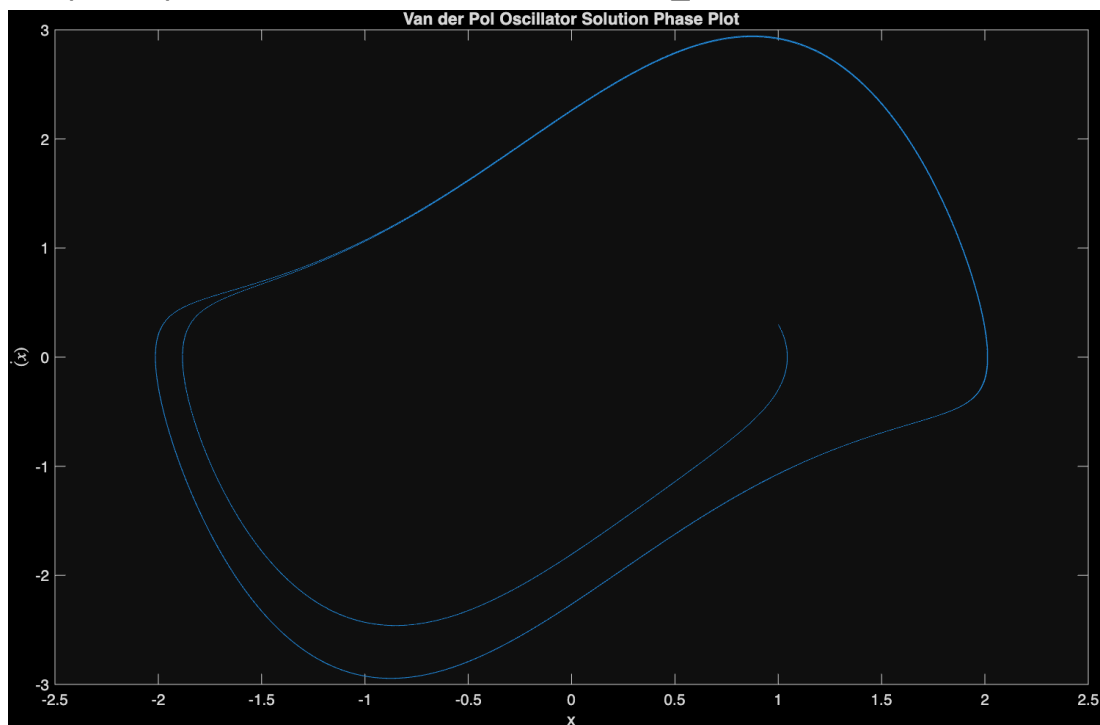
$\boxed{\dot{x}_2 = \mu(1 - x_1^2)x_2 + x_1}$

*correction in figure: $(-x_1)$ in last equation.

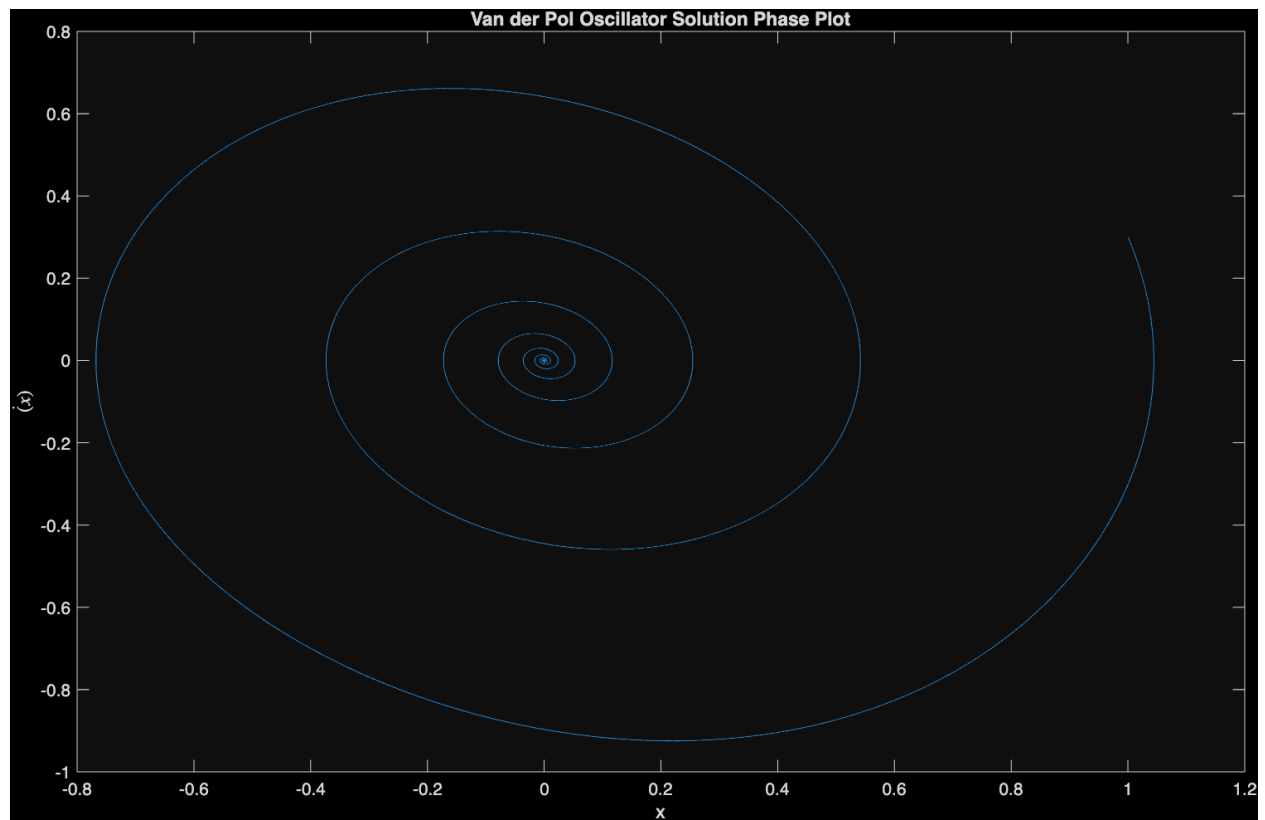
The phase plot of the solution with $\mu = 0.25$, $t_{\text{fin}} = 100$, $x_0 = 1$ looks like this:



The phase plot of the solution with $\mu = 1.25$, $t_{\text{fin}} = 100$, $x_0 = 1$ looks like this:



The phase plot of the solution with $\mu = -0.25$, $t_{\text{fin}} = 100$, $x_0 = 1$ looks like this:

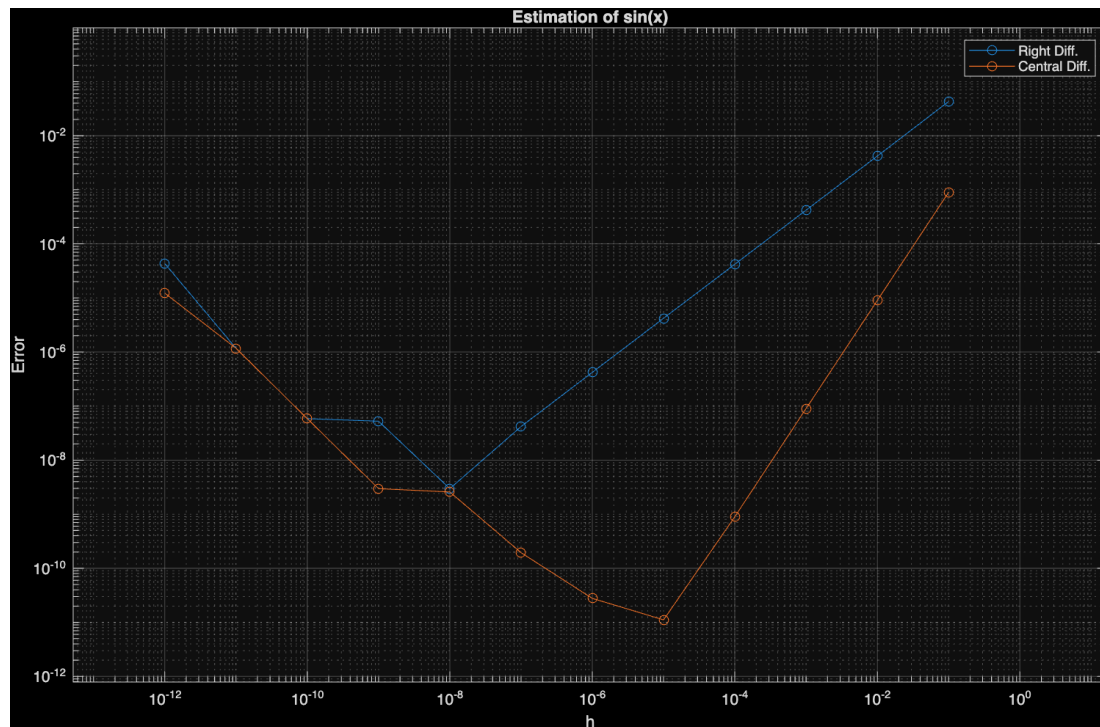


Task 6 - Higher Order Methods

Task 6.a

Calculate the derivative of $\sin(x)$ using central difference. Make a log log plot of error vs h . On one plot, show Euler's method and the central difference method.

Done in file *sin_error.m*



Lowest error with in right difference method with $h = 10e-8$

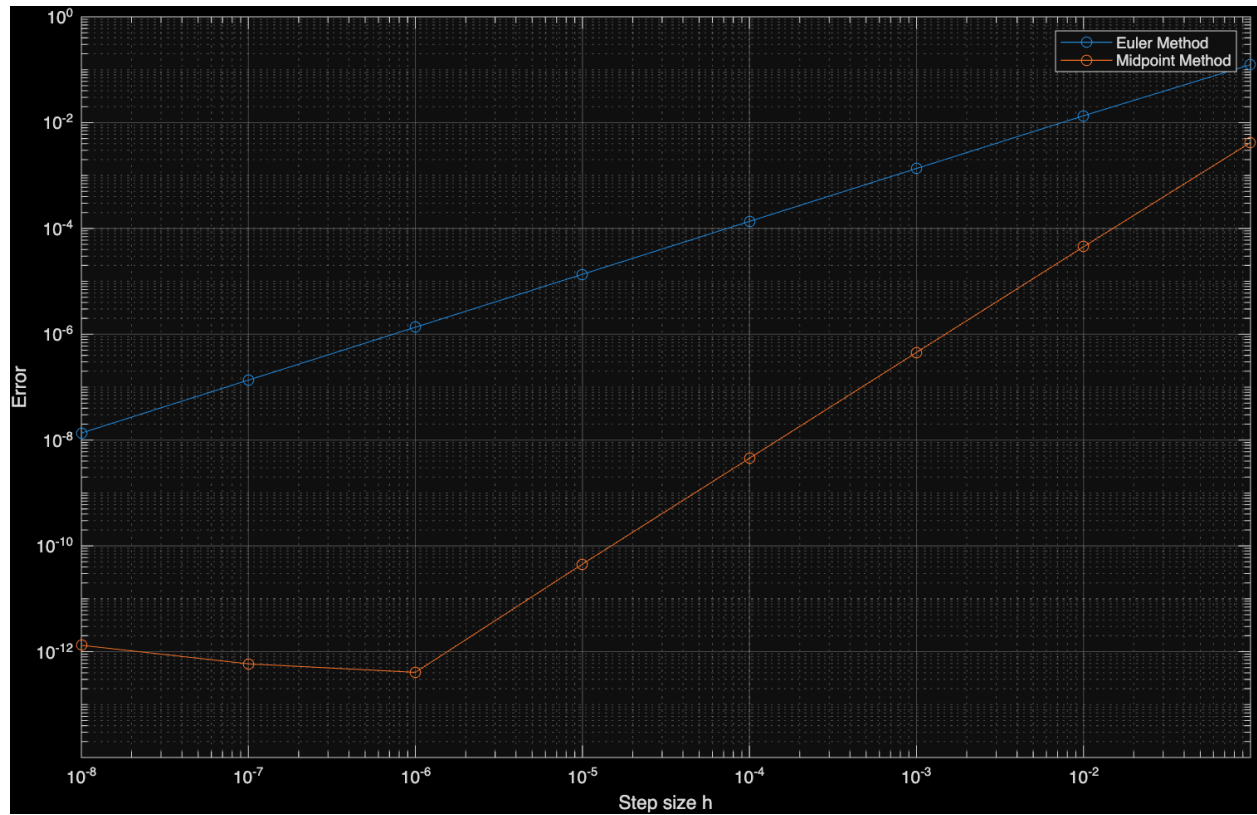
Lowest error with in central difference method with $h = 10e-5$

In the central difference method, the method error goes down with a slope of about twice that of the right difference method.

Task 6.b

Write a new Matlab function that solves arbitrary ODEs using 'the midpoint method'. This is only a slight modification of the Euler solver. Test it with the differential equation $\dot{x} = x$ with initial condition $x(0) = 1$, evaluated at $t = 1$. On one log log plot show the error as a function of h for both the Euler method and the midpoint method.

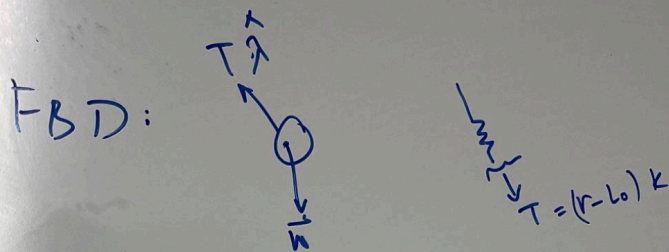
Done in file *ErrorAnalysis.m*



We can also see the error increase again in midpoint method after $h = 10e-6$.

Task 7 - Spring and Mass (2D)

Task 7.a - Find EoM



$$\sum \vec{F} = m\vec{a}$$

$$\vec{w} + T\hat{\lambda} = m\ddot{x}\hat{i} + \ddot{y}\hat{j}$$

$$\frac{1}{m} \left\{ -mg\hat{j} + T \left(\frac{-x\hat{i} + y\hat{j}}{\sqrt{x^2 + y^2}} \right) \right\} = \ddot{x}\hat{i} + \ddot{y}\hat{j}$$

dot w/ $\hat{i}$ and $\hat{j}$,

$$\frac{1}{m} (r - l_0)k \left(\frac{-x}{r} \right) = \ddot{x} \quad \rightarrow \quad \frac{(r - l_0)k}{m} \left(\frac{-x}{r} \right) = \ddot{x}$$

$$-\frac{1}{m} mg + \frac{1}{m} (r - l_0)k \left(\frac{y}{r} \right) = \ddot{y} \Rightarrow -g + \underbrace{\frac{(r - l_0)k}{m} \left(\frac{y}{r} \right)}_{\beta} = \ddot{y}$$

$$z = \text{states} = [x, \dot{x}, y, \dot{y}]$$

$$\dot{z} = [\dot{x}, \ddot{x}, \dot{y}, \ddot{y}]$$

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{y} \\ \ddot{y} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \alpha & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \beta & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ -g \end{bmatrix}$$

Task 7.b - Plot trajectory + animate

*Done in file **SpringMass.m***